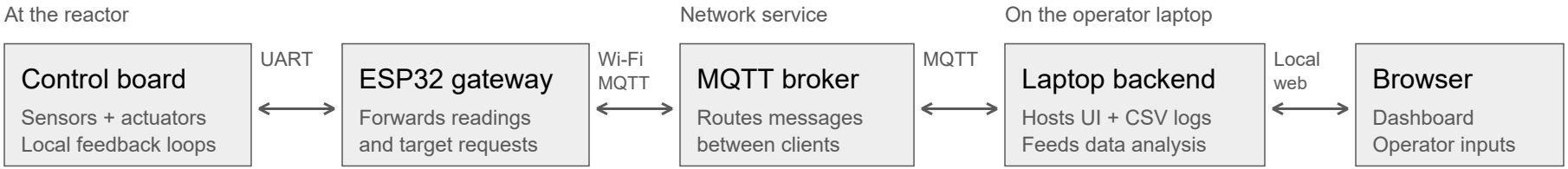
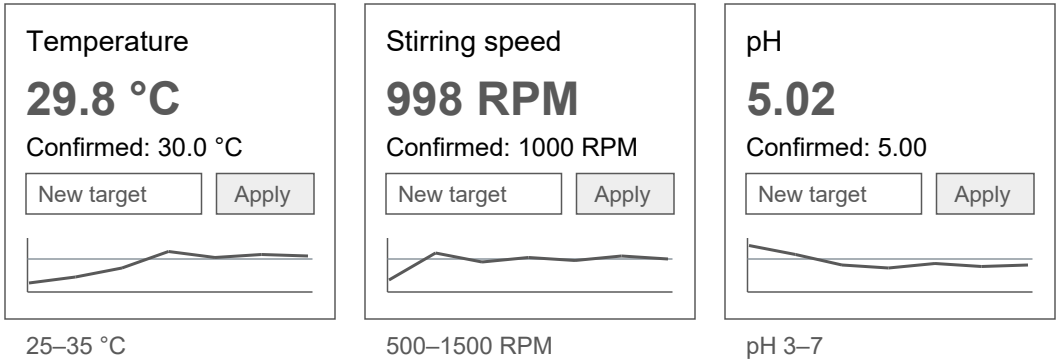
Connectivity and User Interface



Proposed screen (example values)



25–35 °C | 500–1500 RPM | pH 3–7

Operator actions

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

On link loss: mark readings stale.
Local loops follow the agreed policy.

Tests (proposed) 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

Proposed architecture. Verify board models and course-broker access. Brief: Design Specification pp. 4, 13, 17–18; Connectivity pp. 18–19.

Connectivity and User Interface

At the reactor

Control board
Sensors + actuators
Local feedback loops

UART

ESP32 gateway
Wi-Fi access point
HTTP API + UART bridge

Wi-Fi / HTTP
JSON requests

On the operator laptop

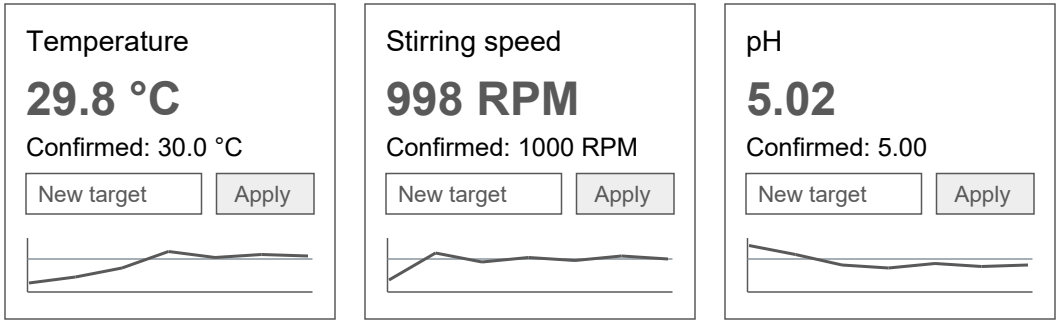
Laptop backend
Polls state + sends targets
Hosts UI + CSV logs

Local web

Browser
Dashboard
Operator inputs

Laptop joins the ESP32 network. Backend polls readings and sends target requests.

Proposed screen (example values)



25–35 °C | 500–1500 RPM | pH 3–7

Operator actions

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

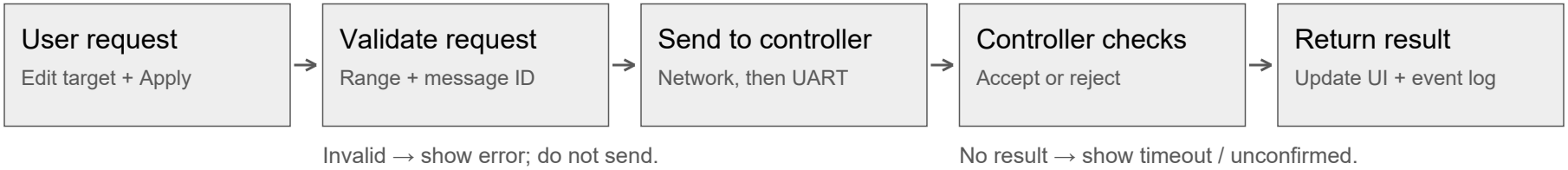
On link loss: mark readings stale.
Local loops follow the agreed policy.

Tests (proposed) 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

Local access only: confirm this scope and access-point use with staff. Internet access is not provided by this arrangement.

Command flow and software structure

Command flow



An applied target is confirmed by the control board. Reaching that target takes longer.

Software structure



First integration test: one simulated reading → dashboard, then one target request → controller confirmation.

Supporting models for Design Specification p. 18. Proposed responsibilities and logic; not measured implementation results.